Aligning LLMs with Biomedical Knowledge using Balanced Fine-Tuning

NEURAL INFORMATION PROCESSING SYSTEMS

Tencent AI for Life Sciences Lab

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Why local confidence matters

Dense low-confidence runs in biomedical text are enriched for rare entities and mechanistic steps. DFT weights every token by absolute confidence and can suppress these knowledge-bearing regions.

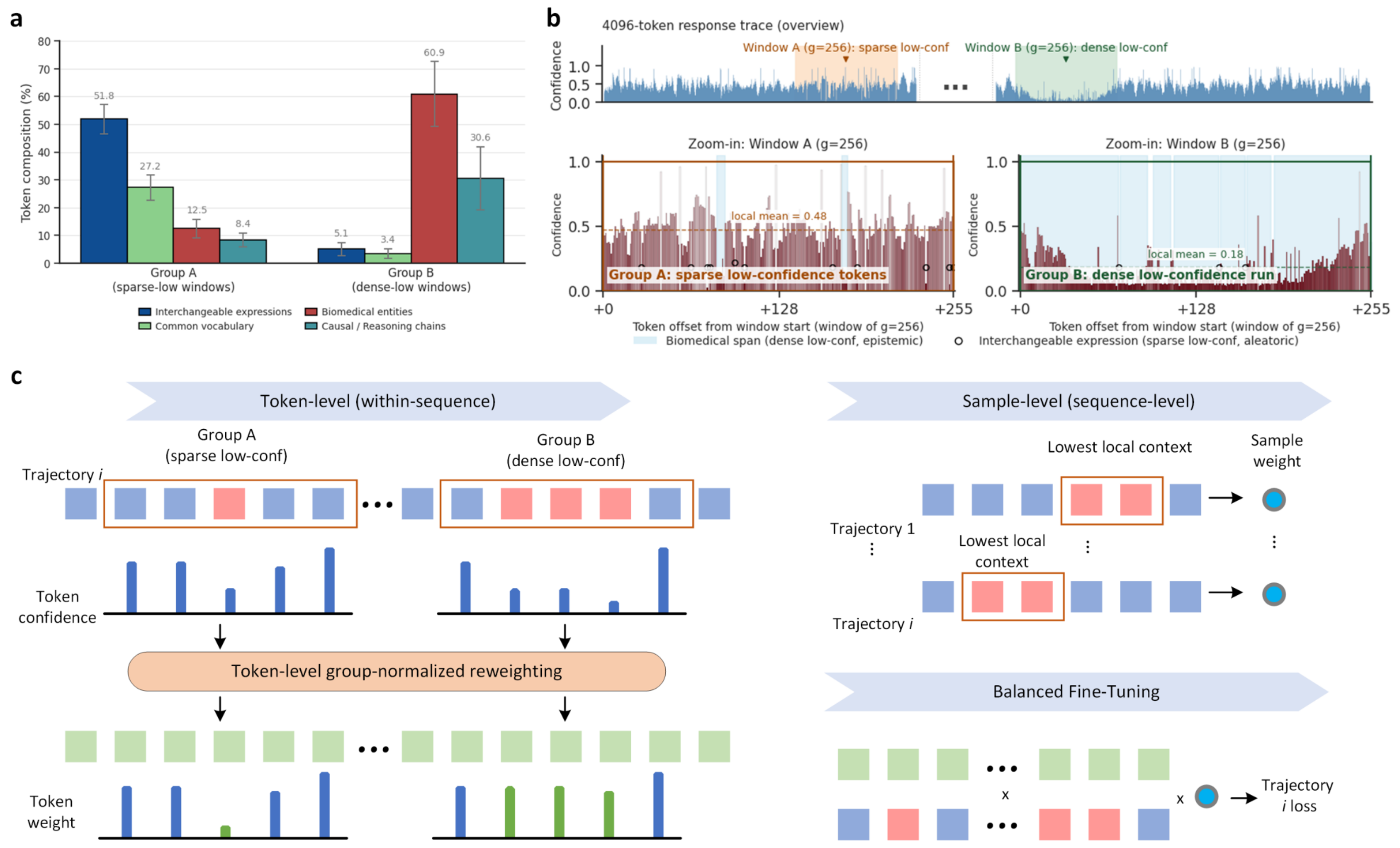

a
Token composition (%)
80
70
60
50
40
30
20
10
0
51.8
27.2
12.5
8.4
5.1
3.4
60.9
30.6
Group A
(sparse-low windows)
Group B
(dense-low windows)
Interchangeable expressions
Common vocabulary
Biomedical entities
Causal / Reasoning chains
b
4096-token response trace (overview)
Window A (g=256): sparse low-conf
Window B (g=256): dense low-conf
Confidence
1.0
0.5
0.0
Zoom-in: Window A (g=256)
Zoom-in: Window B (g=256)
local mean = 0.48
local mean = 0.18
Group A: sparse low-confidence tokens
Group B: dense low-confidence run
+0
+128
+255
Token offset from window start (window of g=256)
Biomedical span (dense low-conf, epistemic)
Interchangeable expression (sparse low-conf, aleatoric)
c
Token-level (within-sequence)
Group A
(sparse low-conf)
Group B
(dense low-conf)
Trajectory i
Token confidence
Token-level group-normalized reweighting
Token weight
Sample-level (sequence-level)
Lowest local context
Sample weight
Trajectory 1
Lowest local context
Trajectory i
Balanced Fine-Tuning
x
Trajectory i loss

Figure 1. Local uncertainty structure and BFT. PMC entity share: 60.9% in dense-low versus 12.5% in sparse-low windows. CRAFT expert spans corroborate concept enrichment (63.8% versus 11.2%).

Balanced Fine-Tuning at two scales

Relative token weights and bounded sample weights

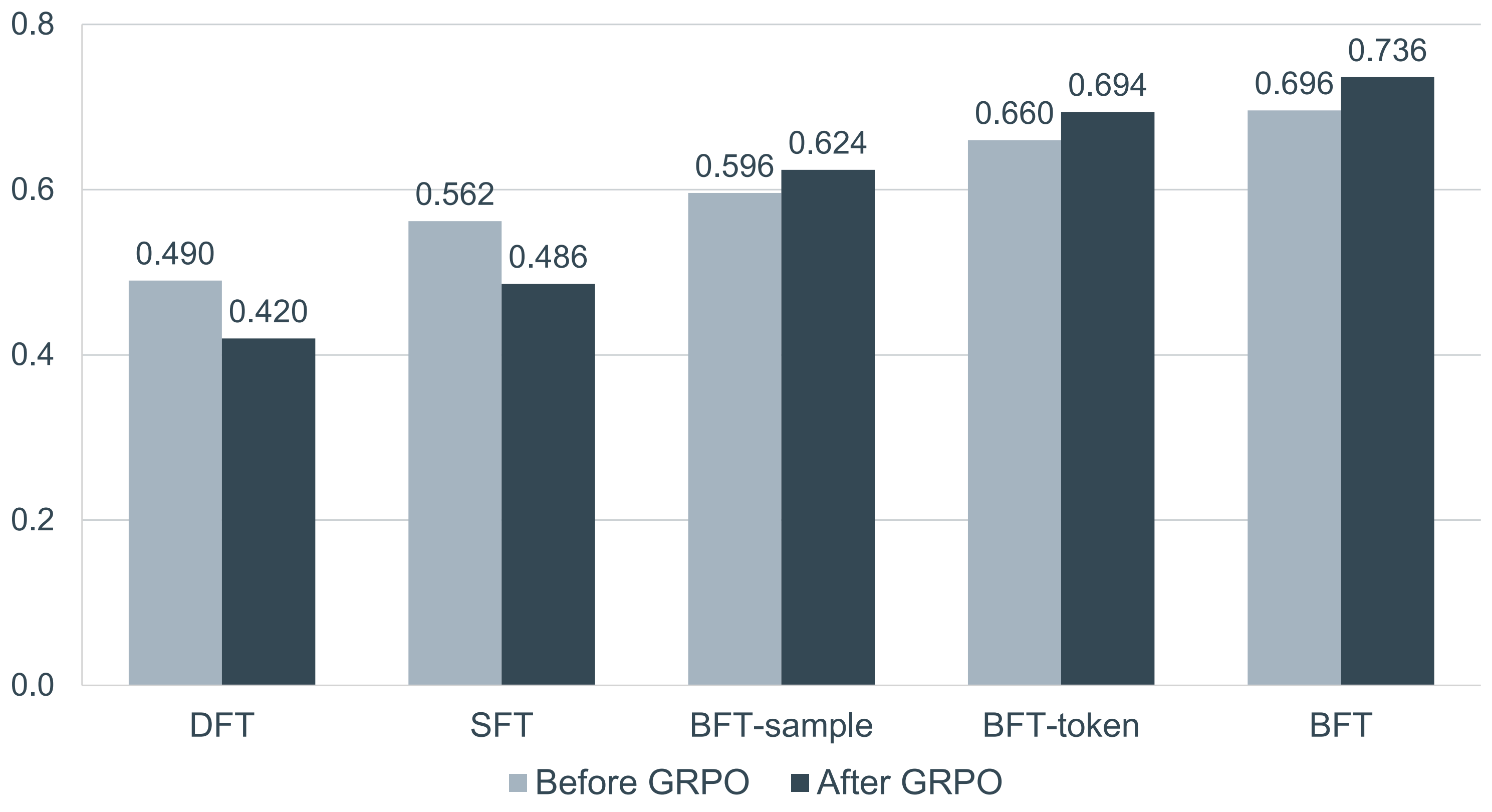
$$\rho_{\beta,t} = \mathrm{clip}(c_{\beta,t} / (C_{\beta,t}^{\mathrm{loc}} + \varepsilon), 0, 1)$$

$$s_\beta = \mathrm{sg}(1 - \min_t C_{\beta,t}^{\mathrm{loc}})$$

The loss combines detached token and sample weights with response-token cross-entropy. The centered window g = 256 is the only additional tunable train-time hyperparameter.

Biomedical reasoning and sparse-reward RL

VCWorld accuracy, 70B backbones
Shared GRPO setup, five held-out cell lines



BFT improves from 0.696 to 0.736 (+4.0 pp) after GRPO, while SFT and DFT decline. Before GRPO, BFT exceeds SFT by 13.4 pp. Its difference from Gemini-2.5-Flash is not significant (p = 0.088).

Medical competence and biological reasoning

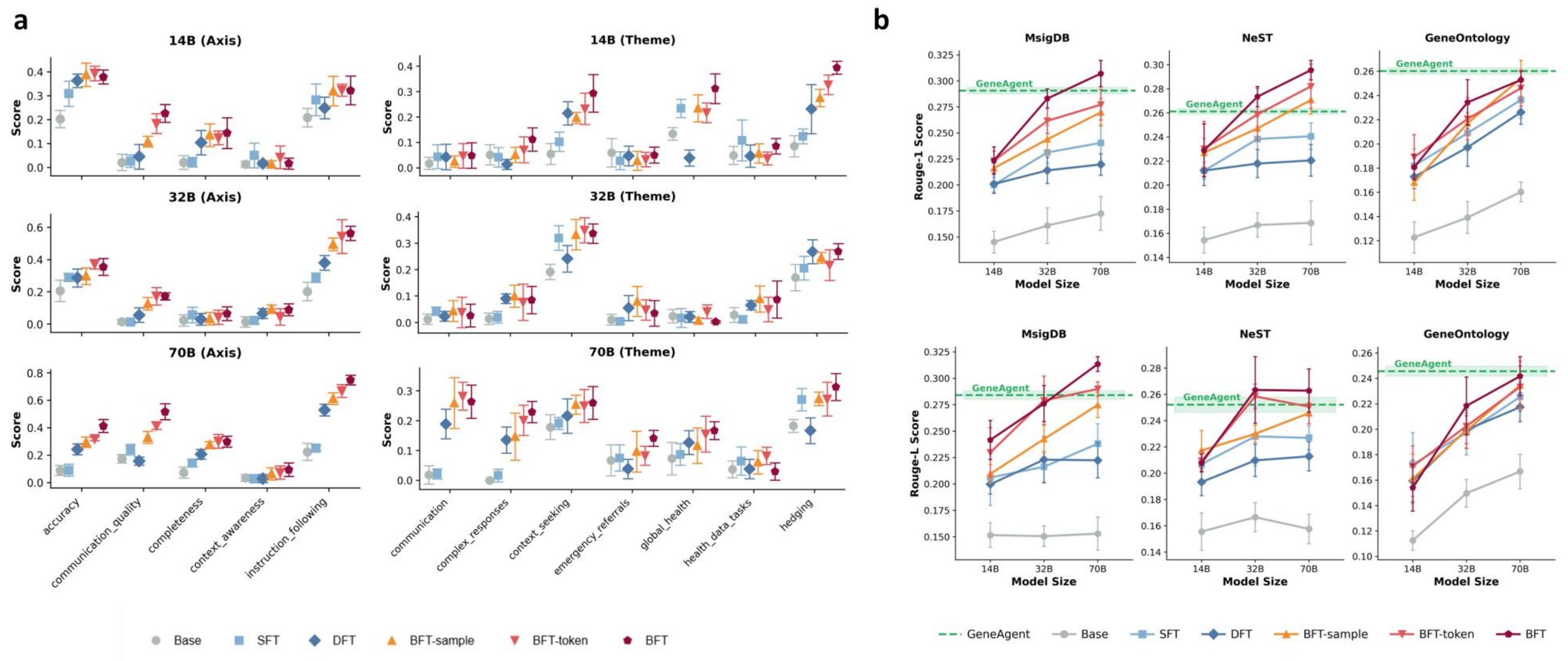


Figure 2. HealthBench Hard and GeneAgent results across 14B, 32B and 70B under a shared supervised training recipe. Error bars quantify decoding variability, not training-seed variability.

Biological representations from generated profiles

BFT-generated profiles are encoded with text-embedding-ada-002.

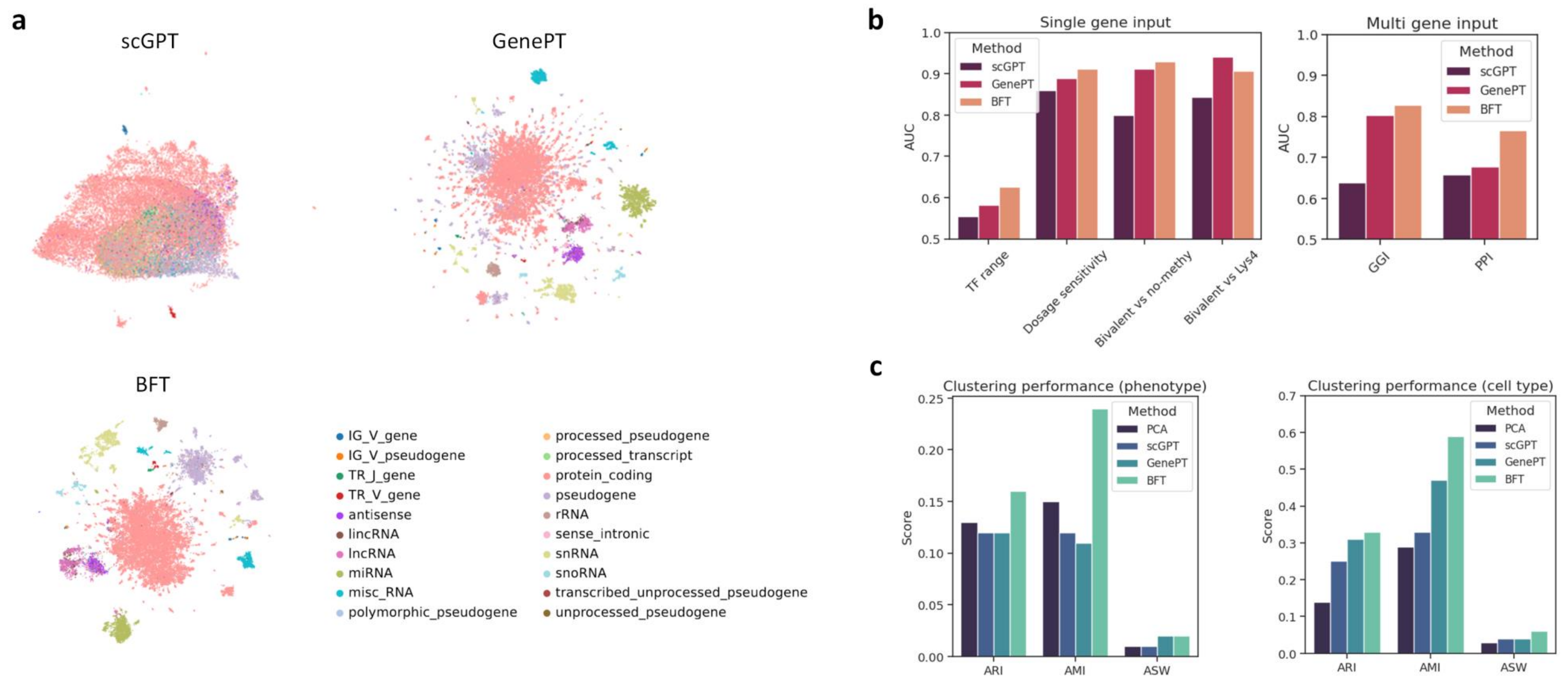
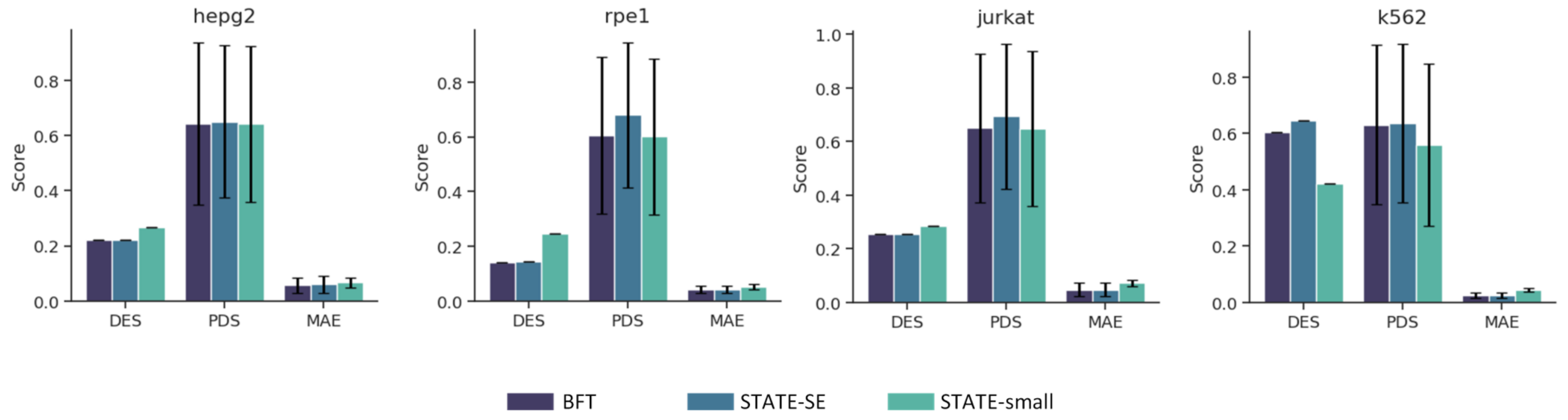


Figure 4. Best average gene-level performance and improved cell clustering on evaluated tasks. GenePT remains higher on Lys4.

Perturbation-response prediction



With the STATE decoder retained, BFT embeddings perform comparably to standard inputs across four datasets. Profile generation is zero-shot for these datasets. Decoder training is not.

Paper and code

arxiv.org/abs/2511.21075
github.com/TencentAILabHealthcare/BFT
Project page: shapsider.github.io/BFT